



**Reflecta**

FACHKONZEPT · NOSQL-ONE-TABLE-DESIGN

# Fachkonzept des NoSQL-One-Table- Designs

ER-Modell, Entitäten, Sekundärindizes, und Access Patterns auf  
Basis von Amazon DynamoDB

PROJEKT

Reflecta

DATUM

26.05.2026

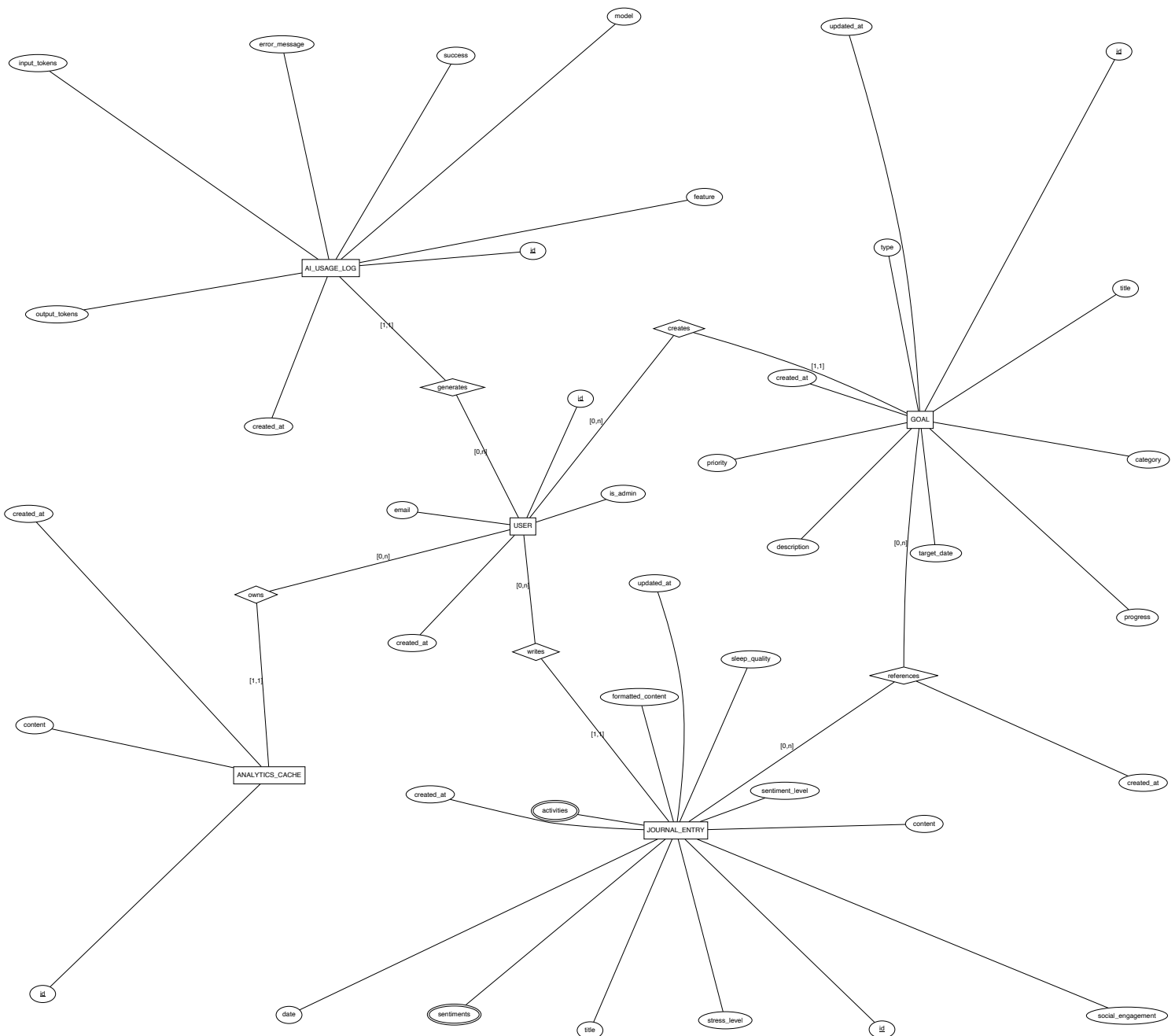
DEPARTMENT

Datenmodell

MITARBEITER

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# 1. ER-Modell mit Kardinalitäten



## 2. Entitäten & Attribute

| ENTITY          | PK         | SK                            | ATTRIBUTES                                                                                                                                               |
|-----------------|------------|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| User            | USER       | <sub>                         | email, is_admin, created_at                                                                                                                              |
| JournalEntry    | USER#<sub> | ENTRY#<entryId>               | title, date, content, sentiment_level, sleep_quality, stress_level, social_engagement, formatted_content, activities, sentiments, created_at, updated_at |
| Goal            | USER#<sub> | GOAL#<goalId>                 | title, type, target_date, category, priority, description, progress, created_at, updated_at                                                              |
| Entry→Goal Link | USER#<sub> | ENTRY#<entryId>#GOAL#<goalId> | created_at                                                                                                                                               |
| Goal→Entry Link | USER#<sub> | GOAL#<goalId>#ENTRY#<entryId> | created_at                                                                                                                                               |
| AnalyticsCache  | USER#<sub> | CACHE#<cache_type>#<period>   | content, created_at                                                                                                                                      |
| AIUsageLog      | USER#<sub> | AILOG#<created_at>#<logId>    | feature, model, input_tokens, output_tokens, success, error_message, created_at                                                                          |

# 3. Secondary Indices

## Local Secondary Indices

### LSI1 — Entries by date (per user)

| ATTRIBUTE | VALUE             |
|-----------|-------------------|
| pk        | USER#<sub>        |
| LSI1_sk   | ENTRY#DATE#<date> |

### LSI2 — Goals by category (per user)

| ATTRIBUTE | VALUE               |
|-----------|---------------------|
| pk        | USER#<sub>          |
| LSI2_sk   | GOAL#CAT#<category> |

### LSI3 — Goals by priority (per user)

| ATTRIBUTE | VALUE                |
|-----------|----------------------|
| pk        | USER#<sub>           |
| LSI3_sk   | GOAL#PRIO#<priority> |

## Global Secondary Indices

### GSI1 — User by email (User Login)

| ATTRIBUTE | VALUE         |
|-----------|---------------|
| GSI1_pk   | EMAIL#<email> |
| GSI1_sk   | USER#<sub>    |

### GSI2 — Admin / cross-user lookups

| ATTRIBUTE | VALUE             |
|-----------|-------------------|
| GSI2_pk   | ENTITY#<type>     |
| GSI2_sk   | <created_at>#<id> |

### GSI3 — AI usage analytics (cross-user)

| ATTRIBUTE | VALUE                |
|-----------|----------------------|
| GSI3_pk   | AILOG#<feature>      |
| GSI3_sk   | <created_at>#<logId> |

## 4. Query-Patterns

### User

| ACCESS PATTERN                          | QUERY                                                |
|-----------------------------------------|------------------------------------------------------|
| Get user profile by sub                 | pk = USER, sk = <sub>                                |
| Get user by email (login)               | GS1_pk = EMAIL#<email>                               |
| List all users                          | pk = USER                                            |
| Admin: List users created in date range | GS12_pk = ENTITY#USER, GS12_sk BETWEEN <t1> AND <t2> |

### JournalEntry

| ACCESS PATTERN                            | QUERY                                                               |
|-------------------------------------------|---------------------------------------------------------------------|
| Get one journal entry                     | pk = USER#<sub>, sk = ENTRY#<entryId>                               |
| List all entries of a user                | pk = USER#<sub>, begins_with(sk, "ENTRY#")                          |
| List entries of a user, sorted by date    | pk = USER#<sub>, begins_with(LS1_sk, "ENTRY#DATE#")                 |
| List entries of a user in a date range    | pk = USER#<sub>, LS1_sk BETWEEN ENTRY#DATE#<d1> AND ENTRY#DATE#<d2> |
| Admin: List all entries                   | GS12_pk = ENTITY#ENTRY                                              |
| Admin: List entries created in date range | GS12_pk = ENTITY#ENTRY, GS12_sk BETWEEN <t1> AND <t2>               |

### Goal

| ACCESS PATTERN                                  | QUERY                                                |
|-------------------------------------------------|------------------------------------------------------|
| Get one goal                                    | pk = USER#<sub>, sk = GOAL#<goalId>                  |
| List all goals of a user                        | pk = USER#<sub>, begins_with(sk, "GOAL#")            |
| List goals of a user by category                | pk = USER#<sub>, LS12_sk = GOAL#CAT#<category>       |
| List all categorized goals (sorted by category) | pk = USER#<sub>, begins_with(LS12_sk, "GOAL#CAT#")   |
| List goals of a user by priority                | pk = USER#<sub>, LS13_sk = GOAL#PRIO#<priority>      |
| List all goals sorted by priority               | pk = USER#<sub>, begins_with(LS13_sk, "GOAL#PRIO#")  |
| Admin: List all goals                           | GS12_pk = ENTITY#GOAL                                |
| Admin: List goals created in date range         | GS12_pk = ENTITY#GOAL, GS12_sk BETWEEN <t1> AND <t2> |

## Entry ↔ Goal Links

| ACCESS PATTERN                              | QUERY                                                     |
|---------------------------------------------|-----------------------------------------------------------|
| List goals linked to one entry              | pk = USER#<sub>, begins_with(sk, "ENTRY#<entryId>#GOAL#") |
| List entries linked to one goal             | pk = USER#<sub>, begins_with(sk, "GOAL#<goalId>#ENTRY#")  |
| Check if specific entry/goal pair is linked | pk = USER#<sub>, sk = ENTRY#<entryId>#GOAL#<goalId>       |

## AnalyticsCache

| ACCESS PATTERN                                  | QUERY                                                   |
|-------------------------------------------------|---------------------------------------------------------|
| Get cached analytics for a user (type + period) | pk = USER#<sub>, sk = CACHE#<cache_type>#<period>       |
| List all caches of a user                       | pk = USER#<sub>, begins_with(sk, "CACHE#")              |
| List all caches of a specific type for a user   | pk = USER#<sub>, begins_with(sk, "CACHE#<cache_type>#") |

## AIUsageLog

| ACCESS PATTERN                                                  | QUERY                                                    |
|-----------------------------------------------------------------|----------------------------------------------------------|
| List all AI usage logs of a user                                | pk = USER#<sub>, begins_with(sk, "AILOG#")               |
| List AI usage logs of a user in time range                      | pk = USER#<sub>, sk BETWEEN AILOG#<t1> AND AILOG#<t2>    |
| Admin: List AI usage logs by feature (cross-user)               | GSI3_pk = AILOG#<feature>                                |
| Admin: List AI usage logs by feature in time range (cross-user) | GSI3_pk = AILOG#<feature>, GSI3_sk BETWEEN <t1> AND <t2> |